Session 1 → Focused on dataset basics: create, view, check missing values.
Session 2 → Introduced data **sources** (CSV, Excel, JSON, SQL, APIs) and integrity checks.
Session 3 → Goes deeper:

- Learn **types of data** (numerical, categorical, text, temporal).
- Understand that sometimes pandas guesses wrong (dates as text).
- Practice **fixing data types**.
- Learn how to **handle categorical variables** (unique values, convert to category).
- Learn **encoding** (label encoding & one-hot encoding).
- Build foundation for **machine learning preprocessing**.

Session – 03 (Identifying Data Types)

Part A – Setup

Q1. Import the pandas library as pd.

Q2. Create a DataFrame using the following data:

Student_ID	Name	Attendance	Final_Grade	Enrolled_Date
1	Alice	85	A	2024-01-10
2	Bob	90	B	2024-02-15
3	Charlie	78	C	2024-01-25
4	David	92	A	2024-03-01
5	Eva	88	B	2024-01-20

Q3. Print the DataFrame.

Part B – Identify Data Types

Q4. Print all column names and their data types.

Q5. Which column looks like a date but is stored as text (object)?

Part C – Convert Data Types

Q6. Convert the Enrolled_Date column into datetime type.

Q7. Print the data types again to confirm the conversion.

Part D – Work with Categorical Variables

Q8. Print all unique values in the Final_Grade column.

Q9. Convert the Final_Grade column into a categorical type.

Part E – Encoding Categorical Data

Q10. Encode the Final_Grade column using label encoding.
(Hint: use .cat.codes)

Q11. Apply One-Hot Encoding to the Final_Grade column.
(Hint: use pd.get_dummies())

Part F – Reflection Questions

Q12. What is the difference between numerical and categorical data?

Q13. Why should we convert date columns into datetime type?

Q14. Why do we need to encode categorical variables?

Extra Practice (for fast finishers)

- Q15. Extract the year from the Enrolled_Date column.
- Q16. Count how many students got grade "A".
- Q17. Find the student with the highest Attendance.